

AJAY MALAH

Java Backend Developer

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Notice Period: Currently Serving

PROFESSIONAL SUMMARY

Java Backend Developer with 3 years of experience (including 1 year internship) building production-grade, distributed systems on Spring Boot. Led the migration of a monolithic railway IoT platform to a 14-service microservices architecture serving Indian Railways, and engineered a high-throughput social platform backend using CQRS/Event Sourcing with the Axon Framework. Strong hands-on expertise in Java, Spring Boot, Spring Security, RESTful APIs, Apache Kafka, Keycloak, PostgreSQL, MongoDB, Redis, Docker, and Kubernetes, with a track record of improving API latency and throughput through caching, query optimization, and unit testing (JUnit, Mockito). Also proficient in Angular and TypeScript for building responsive, component-driven frontend interfaces that integrate with Spring Boot REST APIs. Experienced in Agile/Scrum development, Git-based version control, and CI/CD pipelines. Currently serving notice period and available to join immediately upon completion. Passionate about clean, event-driven system design and building scalable Java backends in production.

TECHNICAL SKILLS

Languages:	Java (primary), SQL, TypeScript
Frontend:	Angular, RxJS, HTML5, CSS3, TypeScript
Core Java & Frameworks:	Spring Boot, Spring Security, Spring Data JPA, Hibernate, Axon Framework (CQRS/Event Sourcing)
Architecture & Patterns:	Microservices, CQRS, Event Sourcing, Saga Orchestration, REST APIs, WebSocket (STOMP), SOLID Principles, Design Patterns
Testing & Build Tools:	JUnit, Mockito, Unit Testing, Integration Testing, Maven, Gradle
Messaging & Streaming:	Apache Kafka, RabbitMQ (AMQP + MQTT plugin)
Databases:	PostgreSQL, MongoDB, Redis (Caching + Pub/Sub), Elasticsearch
Auth & Security:	Keycloak (OIDC/OAuth2), JWT, Policy-Based Access Control (PBAC), Custom AOP Interceptors
DevOps & Infrastructure:	Git, GitHub, Docker, Kubernetes (K3s on VPS), GitHub Actions CI/CD, Jib, Consul, Eureka
Methodology:	Agile/Scrum, Rapid/Iterative Development, Code Reviews, OOP, Data Structures & Algorithms, Multithreading & Concurrency

WORK EXPERIENCE

Software Engineer (Java Backend)

July 2024 - Present

Robokriti India Private Limited — Bengaluru, India (Remote)

- Architected and led the migration of a legacy monolithic railway IoT management & monitoring platform to a 14-service Java/Spring Boot microservice architecture with Consul service discovery, Keycloak SSO, and RabbitMQ event bus — eliminating deployment coupling and enabling per-subsystem independent scaling.
- Designed and delivered 5 new IoT monitoring & management microservices in Spring Boot for Indian Railways — water level monitoring (BR-WLMS, TR-WLMS, Tank-WLMS), trackside environment monitoring (RailTaap), and vibration/voltage management (VVM601/Saathi) — with real-time MQTT telemetry ingestion, threshold alerting, and remote configuration.
- Reduced average database query latency by 85% and improved API throughput by 60% through Redis read-through caching, composite indexing, materialized views, and query plan optimization across PostgreSQL and MongoDB data stores.
- Built a dynamic Policy-Based Access Control (PBAC) engine in Java with custom @RequiresPermission AOP interceptors, runtime permission evaluation via PostgreSQL-backed permission tables, and Redis-invalidated cache with event-driven cache busting — replacing static Keycloak roles with fine-grained resource-level authorization.
- Established a GitHub Actions CI/CD pipeline for the Spring Boot microservices monorepo — detecting changed services automatically and pushing Docker images to GHCR using Jib, significantly reducing release overhead and manual intervention.
- Built an Angular-based internal admin dashboard for railway operations staff to monitor live sensor data, manage device configuration, and view threshold alerts across all 14 microservices.
- Handled 10,000+ requests/hour across live railway monitoring subsystems, maintaining 99.9% system uptime in production environments.

IT Intern - Software Engineer

June 2023 - June 2024

Robokriti India Private Limited — Jabalpur, India

- Contributed to backend maintenance and feature development on the core RailKriti IoT management & monitoring platform, including the MQTT device telemetry pipeline, supporting production deployments on Indian Railways.
- Developed backend telemetry integration modules connecting MQTT device data to core services, and assisted in Keycloak realm configuration, role management, and NDA-compliant access control policies across multiple internal monitoring platforms.

PROJECTS

RailKriti — Railway IoT Management & Monitoring System

2023 - July 2026

Spring Boot · RabbitMQ · MQTT · Keycloak · PostgreSQL · Redis · Angular · Docker · K8s

- Monolith Migration: Led architectural migration of a tightly coupled legacy monolith to 14 independent Spring Boot microservices with Consul service mesh, RabbitMQ event bus, and Keycloak API Gateway — enabling per-subsystem scaling and zero-downtime deployments.
- Monitoring Services: Introduced 5 new monitoring & management microservices for Indian Railways sensor subsystems — water level monitoring (BR-WLMS, TR-WLMS, Tank-WLMS), environment/temperature (RailTaap), and vibration/voltage management (VVM601/Saathi) with real-time MQTT ingestion and threshold alerting.
- Authorization Engine: Designed dynamic PBAC engine with custom AOP interceptors and PostgreSQL permission model — replacing static Keycloak roles with runtime, resource-level permission evaluation and Redis-cached invalidation.
- Admin Dashboard: Built an Angular-based monitoring dashboard consuming the microservices' REST APIs — real-time sensor status, threshold alerts, and remote configuration screens for railway operations staff.

GuppsApp — Distributed Social Media Platform

2024 - Present

Java · Spring Boot · Axon Framework (CQRS/Event Sourcing) · Kafka · PostgreSQL · Redis · Elasticsearch · Angular · K3s/VPS

- Architecture: Designed a cloud-native social media backend with Spring Boot microservices, CQRS/Event Sourcing via Axon Framework, Kafka event streaming, Eureka service discovery, and API Gateway — services cover user, post, feed, comment, like, media, story, DM, and search domains.
- Feed Engine: Built a high-throughput feed service using Redis-based sliding-window velocity scoring and a scheduled scoring job — delivering personalized, ranked feeds with sub-50ms p99 read latency.
- Real-time DM: Implemented WebSocket-based DM service using STOMP over Spring Boot, PostgreSQL-backed message persistence, and Axon @EventHandler delivery receipts.
- Search: Developed search-service on Elasticsearch with Axon event-handler projections — enabling full-text search across users, posts, and hashtags with cursor-based pagination.
- Infrastructure: Provisioned a VPS cluster and deployed the full stack on K3s Kubernetes — Kafka, Axon Server, Keycloak, PostgreSQL, Redis, MinIO, and Elasticsearch on a shared production cluster.
- Web Client: Built an Angular single-page web client with RxJS-driven state management, consuming the platform's REST and WebSocket APIs for feed, DM, and notification features.

EDUCATION

B.Tech - Computer Science & Engineering

2020 - 2024

Rajiv Gandhi Proudyogiki Vishwavidyalaya (RGPV) — Bhopal, India | CGPA: 8.83 / 10.0

CERTIFICATIONS

- Master Microservices with Spring Boot, Docker, Kubernetes — Udemy (Madan Reddy, Eazy Bytes)
- Event Driven Microservices with CQRS, Saga, Event Sourcing — Udemy (Madan Reddy, Eazy Bytes)
- Angular - The Complete Guide — Udemy (Maximilian Schwarzmüller)
- Flutter Advanced: Multi-Modular & Clean Architecture — Udemy (Mina Farid)